

Lirias Arcade

Install Guide

Written for complete beginners -- no coding experience required.

A gift from Liria.

Welcome

This guide walks you through setting up Lirias Arcade from a completely empty computer -- no software installed, no prior experience with Python, Node.js, or RetroArch assumed. Follow the sections in order and you will have a working arcade cabinet front-end by the end.

Sections in this guide:

- 1. Installing prerequisites (Node.js)
- 2. Installing RetroArch and emulator cores
- 3. A note on game files (ROMs) and BIOS files
- 4. The folder structure -- where your files go
- 5. Modifying the software -- what to expect
- 6. Setting up background music
- 7. Changing the banner image
- 8. Arcade cabinet side art
- 9. License, warranty, and legal disclaimer
- 10. Closing note from Liria

1. Installing Prerequisites

Do I need Python?

No. Lirias Arcade's server is built entirely with Node.js. You do not need to install Python or any Python packages to run this project. The only runtime you need is Node.js, covered below.

Step 1 -- Install Node.js

Node.js is the free program that runs the Lirias Arcade web server on your computer.

1. Open your web browser and go to nodejs.org.
2. Click the button offering the "LTS" (Long Term Support) version -- this is the stable, recommended version. Do not pick the "Current" version.
3. The site will download an installer file (it will look like `node-v20.x.x-x64.msi` or similar). Once it finishes downloading, double-click it to open the installer.
4. Click "Next" through the installer screens, accept the license agreement, and leave all options at their default settings. Click "Install", then "Finish" when it's done.
5. Restart your computer. This ensures Node.js is fully recognized by Windows.

Step 2 -- Verify Node.js installed correctly

1. Press the Windows key, type "cmd", and press Enter to open the Command Prompt (a black text window).
2. Type the following and press Enter:

```
node --version
```

If installation succeeded, you'll see a version number such as `v20.11.0`. If you instead see an error like "node is not recognized", restart your computer and try again -- if it still fails, re-run the Node.js installer from Step 1.

You can close the Command Prompt window now.

Step 3 -- First-time setup of Lirias Arcade

The project already includes a one-click launcher that installs its own small set of Node.js dependencies automatically the first time you run it -- you do not need to type any install commands yourself.

1. Open the Lirias-Arcade folder.
2. Double-click `Launch-Arcade.bat`.

The first time you run it, a window will briefly appear installing dependencies (normal -- it happens once, and again only if an update adds a dependency). After that, your web browser will automatically open the game select screen at <http://localhost:8080>.

2. Installing RetroArch and Emulator Cores

RetroArch is the free, open-source program that actually plays your game ROMs (SNES, Genesis, N64, PlayStation, and arcade games). Lirias Arcade is just the pretty menu screen in front of it -- RetroArch does the emulation.

Step 1 -- Download and install RetroArch

1. Go to retroarch.com in your web browser.
2. Click "Download", then choose Windows (and the correct 64-bit / "x86_64" option for most modern PCs).
3. Run the downloaded installer and click through it using the default options.
4. Once installed, note the folder you installed it to (by default this is something like C:\RetroArch-Win64). You'll need this path in Step 3 below.

Step 2 -- Install the required cores

A "core" is a plugin that teaches RetroArch how to emulate one specific game system. Lirias Arcade uses one core per system folder; the core for each folder is listed in `frontend\config.json`. The main ones are:

- FinalBurn Neo (fbneo) -- for the fbneo folder (arcade games)
- MAME2003-Plus (mame2003_plus) -- for the mame folder
- SNES9x (snes9x) -- for the snes folder
- Genesis Plus GX (genesis_plus_gx) -- for the genesis folder
- Mupen64Plus-Next (mupen64plus_next) -- for the n64 folder
- PCSX ReARMed (pcsx_rearmed) -- for the psx folder
- Mesen, Stella, Flycast, Dolphin, LRPS2 and ScummVM -- for the nes, atari, dreamcast, wii, ps2 and PC folders

1. Open RetroArch.
2. From the main menu, go to "Online Updater" (on older versions this may be under "Main Menu" or "Content Downloader").
3. Select "Core Downloader" (sometimes called "Core Updater").
4. One at a time, find and select each core you need and let it download and install. RetroArch will confirm each one once it's finished.

Once downloaded, the core files (with names like `fbneo_libretro.dll`) live inside RetroArch's own "cores" folder, typically something like `C:\RetroArch-Win64\cores`.

Step 3 -- Point Lirias Arcade at your RetroArch install

Open the file `frontend\config.json` inside the Lirias-Arcade folder using Notepad (right-click the file, choose "Open with", then Notepad). Update these two lines to match where you installed RetroArch:

```
"retroarchPath": "E:/RetroArch/retroarch.exe",  
"coresDir": "E:/RetroArch/cores"
```

Replace the example paths with your own install location, save the file (Ctrl+S), and close Notepad. Use forward slashes (/) even though this is Windows -- the example above shows the correct format.

Step 4 -- Copy the included RetroArch settings (optional but recommended)

A starter `retroarch.cfg` file sits in the main Lirias-Arcade folder. It sets up sensible defaults (keyboard controls, gamepad auto-detection for up to 4 controllers, fullscreen) and turns off RetroArch's "loading content" banner, so a game's own intro (for example the SEGA logo) isn't covered by a banner with the game's name. Lirias Arcade applies it automatically each time it launches a game, so you don't need to do anything further -- but if you want these same settings for RetroArch generally, you can copy/merge it into RetroArch's own configuration folder.

3. About Game Files (ROMs) and BIOS Files

No ROMs or BIOS files are included

This repository does not include, provide, distribute, or link to any game ROMs, disc images, or console BIOS files -- for any system, including the PlayStation BIOS files described below.

You are responsible for sourcing any game or BIOS files yourself, from ethical and legal sources only -- for example, by personally dumping/backing up physical games and consoles that you own.

This project does not endorse, recommend, or provide instructions for obtaining copyrighted game or BIOS files from any other source. Downloading copyrighted ROMs or BIOS files you do not own the rights to may be illegal in your country.

It is your responsibility to understand and follow the copyright and emulation laws that apply where you live before adding any files to this project.

PlayStation BIOS files

The PCSX ReARMed core needs a real PlayStation BIOS to run PSX games -- RetroArch cannot emulate a PlayStation without one. The bios folder in this project is included as an empty placeholder (no BIOS files ship inside it). If you own a PlayStation console, you may dump your own BIOS chip and place the resulting files, named exactly as below, into the bios folder:

- scph1001.bin
- scph5501.bin
- scph7001.bin

Dreamcast BIOS files

The Flycast core needs a Dreamcast BIOS to run Dreamcast games. If you own a Dreamcast, you may dump your own and place these two files, named exactly as below, in the bios\dc folder (create it if it is missing). Flycast adds its own memory-card saves and settings files to that folder the first time it runs.

- dc_boot.bin
- dc_flash.bin

This guide does not provide instructions on how or where to obtain these files -- only their expected names and location, for use with a BIOS you have legally sourced yourself. The bios\Mupen64plus folder that ships with this project contains only a public N64 game catalog used by the N64 core (its shader cache is built automatically the first time you play) -- it contains no console firmware and requires nothing further from you.

4. Folder Structure -- Where Your Files Go

Once you have your own legally-obtained game files ready, place each one into the folder matching its system, inside Lirias-Arcade\roms\:

```
Lirias-Arcade/  
  roms/  
    atari/      <- Atari 2600  
    dreamcast/ <- Sega Dreamcast  
    fbneo/      <- FinalBurn Neo (arcade games)  
    genesis/    <- Sega Genesis / Mega Drive  
    mame/       <- Multiple Arcade Machine Emulator  
    n64/        <- Nintendo 64  
    nes/        <- Nintendo Entertainment System  
    PC/         <- ScummVM PC games  
    ps2/        <- PlayStation 2  
    psx/        <- PlayStation  
    snes/       <- Super Nintendo  
    wii/        <- Wii / GameCube
```

The select screen scans these folders automatically every time it loads -- simply drop a game file into the matching folder and refresh your browser (or relaunch the arcade); there's no rebuild or install step required.

5. Modifying This Software

You are welcome to modify, customize, and extend Lirias Arcade to fit your own cabinet, collection, or preferences -- the code is yours to change.

Provided "as-is"

This software is provided "as-is," without warranty of any kind, express or implied. Whether you use it unmodified or make changes of your own, no guarantee is made that it will work correctly, be free of bugs, or be fit for any particular purpose.

No responsibility is claimed or accepted for any issues, damages, data loss, hardware problems, or other consequences that may arise from using or modifying this project. You take on that risk yourself, and any changes you make are your own responsibility.

Personal settings that stay out of the repository

Settings that only make sense on your own machine -- for example a special core for one particular game file -- can go in `frontend\config.local.json` instead of `config.json`. The arcade merges it over `config.json` when it starts, and it is ignored by git, so your personal file names never end up in the shared project.

6. Setting Up Background Music

The game select screen is designed to play a track named `alice.mp3` from the `Lirias-Arcade\music` folder while you browse games. This audio file is not included in this project, due to copyright -- the music folder ships empty.

Option A -- Supply your own `alice.mp3` (recommended, no code changes)

If you have your own audio file that you own the rights to, or that is licensed for this kind of use (royalty-free music, Creative Commons tracks, or something you composed/recorded yourself), you can enable music with no editing required:

1. Rename your audio file to exactly: `alice.mp3`
2. Copy it into the `Lirias-Arcade\music` folder.
3. Relaunch the arcade (or refresh the browser tab) and music will begin playing automatically once you interact with the page.

To rename a file on Windows: right-click the file, choose "Rename", type the new name including ".mp3" at the end, and press Enter. If Windows is hiding file extensions, make sure you don't end up with a double extension like `alice.mp3.mp3` -- the Details view in File Explorer will confirm the file's real name and type.

Option B -- Keep your file's original name (edit the code instead)

If you'd rather not rename your audio file, you can instead tell the code to look for your file's actual name. Open `frontend\index.html` in Notepad and find this line near the top of the file:

```
<audio id="bgm" src="/music/alice.mp3" loop preload="auto"></audio>
```

Change `alice.mp3` to your file's exact name (for example, `my-song.mp3`), save the file, and place that audio file in the `Lirias-Arcade\music` folder. Relaunch the arcade to hear the change.

7. Changing the Banner Image

The background image behind the game select screen is currently `frontend\assets\Liria-Arcade-Banner.jpg`.

Option A -- Replace the file directly (no code changes)

1. Choose the image you'd like to use instead.
2. Rename it to exactly: `Liria-Arcade-Banner.jpg`
3. Copy it into `frontend\assets`, overwriting the existing file when Windows asks.

A wide image (landscape orientation, similar proportions to a widescreen monitor) will look best, since it's used as a full-screen background.

Option B -- Keep your image's original name (edit the code instead)

Open `frontend\style.css` in Notepad and use Find (Ctrl+F) to search for: `Liria-Arcade-Banner.jpg`

You'll find it referenced in two places, each inside a `background-image` style rule, similar to:

```
background-image: url('assets/Liria-Arcade-Banner.jpg');
```

Replace both occurrences with your own file's name (for example, `assets/my-banner.jpg`), save the file, and place your image in `frontend\assets`. Refresh the browser tab to see the new banner.

8. Arcade Cabinet Side Art

Most retro games are squarer than a modern widescreen monitor, so they normally play with black bars on the left and right. Lirias Arcade fills those bars with cabinet-style artwork, the way a real arcade cabinet has decorated side panels around its screen.

How it behaves

- It works for every system -- Genesis, NES, SNES, PlayStation, N64, Dreamcast, PS2, Wii and the rest -- automatically, with nothing to switch on.
- It only ever covers black space. Games that already fill a widescreen monitor (for example 16:9 PS2 or Wii games) show no art at all.
- The artwork is laid on top of the screen like a picture frame. It does not change how the game itself is drawn.
- The very first time you play a particular game, the arcade measures how wide that game's black bars are (a few seconds after it starts) and remembers it. If a game's bars are a different width than expected, it may show plain black bars on that first launch only -- from the next launch on, the art fits exactly.
- Systems whose games can switch between square and widescreen modes while playing (Dreamcast, PS2, Wii) are re-checked every few seconds, and the art appears or disappears to match.

Changing the artwork

The two images are:

```
frontend\assets\left-side.png  
frontend\assets\right-side.png
```

1. Make your own left and right images. A size of 480 x 2160 pixels (tall and narrow) works well on everything from a 1080p to a 4K monitor.
2. Keep the important part of each design away from the far left and right edges. The arcade scales each image to exactly your screen's height (nothing is ever cut off at the top or bottom), centers it in the black bar, and trims a little from both sides when a bar is narrower than the image. When a bar is wider than the image, the extra space beside it simply stays black.
3. Save them as PNG files with exactly the names above, replacing the included ones in frontend\assets.
4. Launch any game. The arcade notices the new images automatically and rebuilds its prepared overlays -- no restart of the arcade is needed.

Turning it off

Open frontend\config.json in Notepad, find the "cabinet" section and change "enabled": true to "enabled": false. Save, then restart the arcade (close it and run Launch-Arcade.bat again). The black bars will be back.

Other settings in the cabinet section

- "screenWidth" / "screenHeight" -- leave at 0 to use your monitor's own resolution. Set them only if your games run on a different screen than your main one.
- "commandPort" -- the local port the arcade uses to talk to RetroArch while a game runs (see below). Change it only if another program already uses 55355.

How the arcade talks to RetroArch

To measure the black bars and switch the art on or off while a game is running, the arcade turns on RetroArch's built-in network command feature for the games it launches, and sends it short commands on this computer only (localhost). Windows may ask once whether RetroArch may use the network. We recommend choosing Cancel (decline): the art still works, because the commands never leave this computer, and declining keeps other devices on your network from sending RetroArch commands.

These settings are added only for games started from the arcade. If you use RetroArch's own menu to "Save Current Configuration" while such a game is running, they would be saved into RetroArch's normal settings too -- that is harmless, but you can undo it by turning off Settings > Network > Network Commands.

Generated overlay images and remembered measurements are stored in frontend\assets\cabinet-cache. It is safe to delete that folder at any time -- it is rebuilt automatically.

9. License, Warranty, and Legal Disclaimer

Lirias Arcade is open-source tooling: a menu/launcher front-end for RetroArch, provided free of charge, with no game content of any kind included.

No warranty

This software is provided "as-is," without warranty of any kind, either expressed or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose. In no event shall the authors or contributors be held liable for any claim, damages, or other liability arising from, out of, or in connection with the software, or its use or other dealings in the software.

Your responsibility

You are solely responsible for ensuring that anything you add to this project -- game ROMs, BIOS files, replacement music, replacement images, or code changes -- complies with the copyright and other laws that apply to you. This project does not provide, host, endorse, or link to copyrighted game content, BIOS files, or unlicensed media of any kind, and takes no responsibility for how you choose to use it.

10. A Note From Liria

Lirias Arcade is offered freely, as a gift from Liria.

If you've enjoyed it and would like to support the Hotel Liria project, donations are always appreciated at:

DeFiFry.eth

Thank you for playing, and enjoy the arcade.